Sampled or Stated? Post-Training Moves a Language Model's Differential Diagnosis from Its Samples to Its List

Jairam Hathwar
University of Washington

Draft, October 24, 2026

Abstract

A language model's differential diagnosis can be read two ways: from the ranked list it states in one answer, or from the spread of its answers over repeated samples. Pass@k, self-consistency and sampling-based uncertainty all use the second. We compare the two readouts under a controlled design: base/post-trained pairs from two open families at 7–32B (Qwen2.5-7B and -32B, Qwen3-8B, and every released stage of OLMo 3 7B and OLMo 3.1 32B) and three medical models fine-tuned from one parent, evaluated with 32 samples per case on DDXPlus, whose patients come with a reference differential, and on rare-disease cases from Phenopacket Store. Post-training consistently moves the useful readout from the samples to the list. The sampled differential collapses: post-trained models trail their base models by 4–28 points at pass@32 and give 1.3–2.6 distinct top answers per 32 samples against 3.8–8.3. Majority voting over 32 samples gains the general post-trained models at most 1.9 points, against 3–8 for their base models, and agreement between samples becomes a weaker error signal (AUROC -0.06 to -0.16). The stated list improves at the same time: in every family and size, a single stated differential moves closer to the reference while the samples move away (for OLMo 3 7B only when asked for a full distribution over all conditions), and the post-trained Qwen models name can't-miss conditions 9–13 points more often. Which of the two is the better error signal depends on the model, but a cheap audit predicts it: across 56 models, 100 cases with five samples each pick the better signal in 88% of draws, losing 0.004 AUROC on average. Which post-training stage narrows the samples most depends on the recipe: the RL stage in OLMo 3 7B, SFT in OLMo 3.1 32B. Neither readout tracks how ambiguous a case is, and on rare disease medical RL raises stated confidence from 0.51 to 0.67–0.71 at about 2% accuracy. Claims about a model's diagnostic breadth should say which readout they use.

1 Introduction

Clinicians reason over a differential diagnosis, a ranked set of conditions consistent with the presentation, and the conditions they keep in view determine which tests are ordered and which dangerous possibilities are ruled out. Language-model evaluations in medicine nonetheless score a single answer: multiple-choice accuracy, top-1 diagnosis, or at most pass@k for a small k. Post-training, whether general instruction tuning and RLHF or domain-specific medical reinforcement learning and distillation, is reported to improve these scores [1, 9, 15].

A model's differential can be read in two ways: from the ranked list it states in one answer, or from the distribution of its answers over repeated samples. Much current practice uses the second. Pass@k, self-consistency voting [23] and sampling-based uncertainty such as semantic entropy [8] all draw on repeated samples, and so does the evidence that reinforcement learning sharpens models, which win at pass@1 while their base models solve more problems given many samples [19, 25, 26].

The two readouts agree for a model that samples from the distribution it states. Work outside medicine suggests that aligned models do not. They can verbalize distributions they fail to sample from [18, 27], their verbalized confidence is better calibrated than their token probabilities [22], and their samples can homogenize enough to defeat sampling-based uncertainty [16]. Whether post-training opens this gap, and what it does to a differential diagnosis, where a reference distribution exists, has not been measured.

We measure it directly. Our design holds the pretrained model fixed and varies only the post-training: (i) two base/instruct pairs, Qwen2.5-7B and Qwen3-8B, which isolate general post-training; (ii) three public medical models fine-tuned from the same parent, Qwen2.5-7B-Instruct: AlphaMed (rule-based RL), HuatuoGPT-o1 (SFT then PPO), and m1 (SFT distillation), plus AlphaMed's companion checkpoint trained from the base model; (iii) every released stage of OLMo 3 7B [21], from one base model through SFT, DPO and RL, plus a checkpoint trained with RL directly from the base; (iv) at 32B, Qwen2.5-32B with its Instruct model and every released stage of OLMo 3.1 32B. We draw 32 samples per case. On DDXPlus [7], which ships a reference differential for every synthetic patient, we score both readouts against that reference; on rare-disease cases from Phenopacket Store [5] we score accuracy and stated confidence.

Findings.

1. **Post-training collapses the sampled differential.** In both Qwen families at 7–8B the post-trained model is far better on its first answer (+13 to +27 points) but its pass@k curve crosses the base model's at k≈4–8, and at k=32 the base model leads by 10–15 points; at 32B the lead is smaller (4.0–4.9 points). Post-trained models give 1.3–2.6 distinct top answers per 32 samples, against 3.8–8.3, and can't-miss conditions drop out of their samples (§4.1, §4.2, §4.5). Which stage narrows the samples most depends on the recipe: the RL stage applied after SFT and DPO in OLMo 3 7B (19 points of pass@32), SFT in OLMo 3.1 32B (12 points), where the RL stage costs only 3 (§4.4).

2. **The stated differential does not collapse; it improves.** A single stated list from the post-trained Qwen models at 7–32B is closer to the reference differential than the base model's (total-variation distance −0.04 to −0.09), covers 6–11 points more of its mass, and names can't-miss conditions 9–13 points more often, while their samples move away from the reference. The same holds for OLMo 3.1 32B. OLMo 3 7B's five-item lists do not improve, but its stated distribution does when asked for a probability on every condition (§4.6, §4.8).

3. **Sampling-based methods lose their value; which signal to trust is model-dependent but cheap to check.** Majority voting over 32 samples gains base models 3–8 points at T=0.7 and the general post-trained models at most 1.9; models that reason before answering still gain 4–5. For post-trained Qwen models, the top three of one stated list beats the three most frequent of 32 sampled answers by 6–14 points. Agreement between samples becomes a weaker error signal in every family; the stated probability overtakes it for Qwen3-8B and OLMo-3-7B-Instruct but not for the other three general post-trained models. Across 56 models the winner is predicted by how often a model's samples all agree and how informative its stated probability is, and a 100-case, five-sample audit picks it in 88% of draws. Neither readout tracks how ambiguous a case truly is (§4.7, §4.9).

4. **Medical post-training adds accuracy, not breadth, and raises overconfidence on rare disease.** AlphaMed improves on its parent at every k without widening its sampled differential; its companion checkpoint trained from the base model keeps most of the base

model's diversity (§4.10). With accuracy near 2%, medical RL raises stated confidence from 0.51 to 0.67–0.71, and every OLMo stage states 0.73–0.84 at about 1% accuracy (§4.11).

5. **RL from the base model need not collapse the samples.** GRPO from Qwen3-8B-Base nearly matches the official post-trained model's top-1 accuracy while keeping more than twice its sampled diversity, and a group-level reward narrows the samples less than standard RLVR at equal accuracy (§5).

2 Related work

What RL does to a model's output distribution. Yue et al. [26] find that RL with verifiable rewards improves pass@1 on math and code while base models win at large k. Wu et al. [25] argue that RLVR mostly reweights solutions already in the base model's support. Oh et al. [19] document a "sharpening tax" across 14 base/post-trained pairs on agentic benchmarks. In contrast, Cheng et al. [3] report that RL expands the boundary on domains under-represented in pretraining. In medicine, the only pass@k analysis we know of studies vision-language models and finds that RL mainly sharpens [10]. All of these measure the sampled distribution; we also measure the stated one.

Diversity loss from alignment. Instruction tuning and RLHF reduce output diversity [12, 24]. Related work explains this as typicality bias in preference data [27] and documents cross-model homogeneity [11]. Zhang et al. [27] also track diversity across the SFT, DPO and RLVR checkpoints of Tülu 3 on creative writing; we do the same for OLMo 3 against a clinical reference distribution.

Stated versus sampled distributions. Meister et al. [18] find that language models asked to verbalize a group's opinion distribution match it better than their samples or log-probabilities do, and that they "struggle to sample from their own distribution"; they study instruct and API models on opinion surveys, which have no correct answer. Verbalized sampling [27] prompts aligned models to state a distribution over candidate responses and recovers part of the diversity lost in post-training. Tian et al. [22] show that RLHF models' verbalized confidence is better calibrated than their token probabilities. Liu [16] report that aligned models' samples often fall into a single semantic cluster, leaving sampling-based uncertainty [8] without signal on those questions. We add a task with a known reference distribution, controlled base/post-trained comparisons in three families showing that post-training opens the gap, and its consequences for diagnostic aggregation and uncertainty.

Medical reasoning models and calibration. Medical post-training recipes include verifier-guided SFT and PPO [1], minimalist rule-based RL [15], and test-time-scaling distillation [9]. Reward models and RL tend to favor confident answers [4, 13], and reasoning models are often overconfident [17]. We show that this overconfidence appears in public medical descendants of one parent, on rare disease.

Differential diagnosis benchmarks and distributional RL. DDXPlus [7] was introduced for automatic diagnosis systems trained on differentials [6]. Recent LLM evaluations on it score hierarchical top-k accuracy [14]. Puri et al. [20] train models with GRPO to output several answers per forward pass, rewarding coverage of a gold *set* and set-level calibration, and evaluate on DDXPlus. We instead compare stated and sampled differentials against the reference *distribution*

Model	Parent	Post-training	Role
Qwen2.5-7B	n/a	none (pretrained)	base
Qwen2.5-7B-Instruct	Qwen2.5-7B	general SFT + RLHF	instruct; medical parent
AlphaMed-7B [15]	Qwen2.5-7B-Instruct	rule-based RL	medical
HuatuoGPT-o1-7B [1]	Qwen2.5-7B-Instruct	SFT + PPO (verifier)	medical
m1-7B-23K [9]	Qwen2.5-7B-Instruct	SFT distillation	medical
Qwen3-8B-Base	n/a	none (pretrained)	base
Qwen3-8B	Qwen3-8B-Base	general post-training	instruct
Qwen2.5-32B	n/a	none (pretrained)	base
Qwen2.5-32B-Instruct	Qwen2.5-32B	general SFT + RLHF	instruct
OLMo-3-1025-7B [21]	n/a	none (pretrained)	base
OLMo-3-7B-Instruct-SFT	OLMo-3-1025-7B	SFT	stage 1
OLMo-3-7B-Instruct-DPO	OLMo-3 SFT	DPO	stage 2
OLMo-3-7B-Instruct	OLMo-3 DPO	RLVR	stage 3 (instruct)
OLMo-3-7B-RL-Zero-General	OLMo-3-1025-7B	RL (LM judge)	RL from base
OLMo-3-1125-32B	n/a	none (pretrained)	base
OLMo-3.1-32B-Instruct-SFT	OLMo-3-1125-32B	SFT	stage 1
OLMo-3.1-32B-Instruct-DPO	OLMo-3.1 SFT	DPO	stage 2
OLMo-3.1-32B-Instruct	OLMo-3.1 DPO	RLVR	stage 3 (instruct)

Table 1: Models studied. Each pair or family shares its pretrained weights, so differences reflect post-training. OLMo 3 and OLMo 3.1 release every stage of their instruct pipelines.

across public post-trained model families. Our group-level reward operates on the distribution of independently sampled answers (§5).

3 Experimental setup

Tasks. **DDXPlus** [7]: synthetic patients produced by a rule-based medical knowledge base. Each patient has a true pathology and a reference differential, a probability distribution over the 49 conditions. We use 980 test patients, 20 per condition. Each patient's coded symptoms and history are rendered as a short vignette; we fix known mistranslations in the English evidence descriptions. The prompt lists all 49 conditions and asks for the five most likely, with probabilities, as JSON. **Phenopacket Store** [5] (v0.1.27): published rare-disease cases annotated with HPO phenotypes and an OMIM diagnosis. We use a disease-balanced set of 1,401 cases covering 749 diseases (at most two per disease). Each case is a vignette of present and absent phenotypes, and the model is asked for an open-ended differential. Answers are graded by mapping names to Mondo; we credit an exact match, and separately a "related" match to a specific ancestor group or subtype.

Models. Table 1. All checkpoints are public. For the medical models we verified the parent model from each released configuration.

Sampling. We draw n=32 samples per case with vLLM, at T=0.7, top-p 0.95, and at T=1.0, top-p 1.0 (sampling from the model's own distribution). All models use their chat template and the same prompt; OLMo-3-1025-7B ships without one, so it gets its SFT checkpoint's template. Completions are capped at 4,096 tokens (8,192 for m1, whose long reasoning otherwise truncates), and generation stops at the end-of-turn token. The 32B models are sampled at T=0.7 only; the OLMo 3.1 32B SFT and DPO checkpoints, stored in float32, are run in bfloat16 like the other

Pair	Model	pass@1	pass@32	Distinct	Entropy	TV↓	Cov@32	ECE↓
Qwen3-8B, T=1	base	0.282	**0.789**	**6.60**	**1.31**	**0.712**	**0.407**	0.214
	post-trained	**0.556**	0.643	1.53	0.18	0.783	0.236	**0.061**
Qwen2.5-7B, T=1	base	0.239	**0.766**	**8.23**	**1.58**	**0.715**	**0.422**	0.303
	instruct	**0.464**	0.663	2.65	0.49	0.761	0.290	**0.137**
Qwen3-8B, T=0.7	base	0.388	**0.730**	**3.76**	**0.78**	**0.750**	**0.322**	0.121
	post-trained	**0.558**	0.607	1.27	0.11	0.793	0.217	**0.063**

Table 2: DDXPlus, 32 samples per case (980 cases). The reference differential has mean entropy 1.87. Distinct and entropy refer to the top-1 answers across samples; Cov@32 is the share of reference probability mass covered by those answers. ECE is for the stated top-1 probability.

stages. We take the last valid JSON list in each completion. Unparsable completions count as incorrect; this affects mainly base models, so base-model pass@k is a lower bound at large k, while post-trained gains at small k are partly format. §4.3 reports format-controlled comparisons.

Metrics. *pass@k*: the unbiased estimator of Chen et al. [2] on the top-1 answer of each sample. *Distinct* and *entropy*: the number of distinct top-1 diagnoses across a case's 32 samples, and the entropy of their empirical distribution, averaged over cases. On DDXPlus, *TV*: total-variation distance between this empirical distribution and the reference differential. *Coverage@k*: the expected share of the reference differential's probability mass covered by the distinct top-1 answers in k random samples. *Calibration*: expected calibration error (10 bins) of the stated top-1 probability against top-1 correctness. These metrics describe the *sampled* readout. §4.6 defines the corresponding metrics for the *stated* readout, the list in a single completion. Differences come with 95% paired-bootstrap intervals over cases (2,000 resamples).

4 Results

4.1 General post-training trades coverage for top-1 accuracy

Table 2 and Figure 1 compare each base model with its post-trained counterpart on DDXPlus. In both families the post-trained model is much better on its first answer: +27.5 points for Qwen3 and +22.5 for Qwen2.5 at T=1 (intervals $[+25.3, +29.6]$ and $[+20.7, +24.4]$). The advantage shrinks with k and reverses: the curves cross at $k\approx$4–8, and at k=32 the base model leads by 14.6 points (Qwen3, $[+12.2, +17.1]$) and 10.3 points (Qwen2.5, $[+7.8, +12.8]$). The pattern is the same at T=0.7 for Qwen3, crossing at k=4 with a base lead of 12.2 points at k=32 ($[+9.9, +14.9]$). Unparsable base-model samples count as wrong, so these gaps are lower bounds. Qwen2.5 at T=0.7 behaves the same way: +12.6 points at k=1 $[+10.7, +14.5]$ and a base lead of 9.7 at k=32 $[+7.3, +12.0]$.

4.2 The sampled differential collapses, while top-1 calibration improves

The gap at large k reflects a collapsed *sampled* differential. Across 32 samples, post-trained models name 1.3–2.6 distinct diagnoses per case, against 3.8–8.3 for their base models. Their answer entropy is 0.11–0.49, against 0.78–1.59 for base models and 1.87 for the reference differential (Table 2). Sampling at T=1 widens the base models' answers but barely changes the post-trained models' (Qwen3 entropy 0.11 at T=0.7, 0.18 at T=1), so the collapse lives in the model's own distribution rather than in the decoding settings. As a result, post-trained models cover less of the

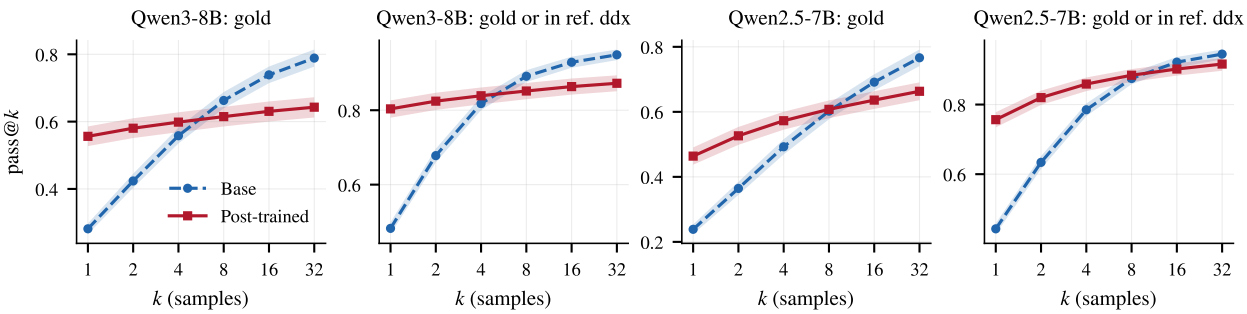


Figure 1: pass@k on DDXPlus at T=1 (980 cases, 32 samples), base model (dashed) vs. its post-trained version (solid), with 95% case-bootstrap bands. "Gold": the top-1 answer is the true pathology; "gold or in ref. ddx": it is the true pathology or appears in the reference differential. In both families the post-trained model wins at small k and loses at large k.

reference differential: 0.24 vs. 0.41 of its probability mass at k=32 for Qwen3, and 0.29 vs. 0.42 for Qwen2.5. Their top-1 calibration, however, is markedly better (ECE 0.06 vs. 0.21 and 0.14 vs. 0.30). On every single-answer criterion (accuracy, small-k pass@k, calibration), the post-trained model therefore looks strictly better, and the loss of breadth in its samples is invisible. Whether the breadth is lost from the model's stated differential as well is a separate question (§4.6).

Which diagnoses drop out? For each case and each condition that the reference differential weights at least 0.1, we ask whether the model names that condition as its top answer in at least one of its 32 samples. We call this *considering* the condition. We group the 49 conditions by DDXPlus severity (1 = most severe). Post-training lowers consideration in every tier (Table 3), and the severe "can't-miss" tier is not spared. For the 17 severity-1–2 conditions (e.g. NSTEMI/STEMI, acute pulmonary edema, anaphylaxis, pulmonary embolism, unstable angina), consideration falls from 0.45 to 0.27 for Qwen3 and from 0.51 to 0.34 for Qwen2.5 at T=1. The differences are -18.4 $[-20.8, -15.9]$ and -17.0 $[-19.7, -14.2]$ points. Some diagnoses disappear from the top answer almost entirely. Qwen2.5-7B-Instruct names unstable angina as its top answer in one of 107 eligible cases, where its base model does in 47%. When unstable angina is the true pathology, pass@32 falls from 1.00 to 0.00. Stable angina (0.89 to 0.06) and myocarditis (0.41 to 0.00) follow the same pattern. These conditions are not forgotten: they still appear lower in the listed differential (unstable angina in 50% of eligible cases, against 95% for the base model). The model ranks them but no longer commits to them. At the same time, post-trained models concentrate on a few salient "attractor" diagnoses. Qwen2.5-7B-Instruct gives NSTEMI/STEMI as its top answer in 13.1% of samples, against a reference rate of 4.4% and 6.4% for its base model. It also over-predicts pulmonary embolism (8.2% vs. 3.2%) and pneumonia (8.0% vs. 2.4%). Chest-pain presentations thus collapse onto the most dangerous candidate. That raises its accuracy (NSTEMI pass@1 0.99 vs. 0.44) at the cost of the alternatives, which is a defensible triage bias but not a differential.

4.3 Robustness to prompt and output format

Prompt. We repeat the Qwen3-8B pair with two other prompts (Table 4). *Plain* drops the system prompt and rewords the instructions. *CoT* asks the model to reason step by step before answering; Qwen3's thinking mode stays off in all runs. The plain prompt reproduces the pattern: the post-trained model gains $+23.0$ $[+20.9, +25.1]$ points at k=1 and loses -7.1 $[-9.3, -5.0]$ at k=32 (T=0.7; at T=1, $+34.2$ and -8.7 $[-11.0, -6.4]$), with 1.3–1.5 distinct answers against 3.7–

Comparison	Considered@32 (base → post-trained) Severe (1–2)	Moderate (3)	Mild (4–5)
Qwen3-8B, T=1	0.45 → 0.27	0.50 → 0.32	0.45 → 0.20
	−18.4 [−20.8, −15.9]	−18.1 [−20.7, −15.7]	−24.2 [−27.0, −21.4]
Qwen2.5-7B, T=1	0.51 → 0.34	0.50 → 0.37	0.38 → 0.23
	−17.0 [−19.7, −14.2]	−12.1 [−14.7, −9.7]	−15.0 [−17.4, −12.6]
Qwen3-8B, T=0.7	0.34 → 0.24	0.44 → 0.29	0.31 → 0.18
	−9.8 [−11.8, −7.9]	−15.8 [−18.2, −13.5]	−13.4 [−15.8, −11.1]
Qwen2.5-7B, T=0.7	0.40 → 0.28	0.41 → 0.32	0.28 → 0.17
	−12.4 [−14.7, −10.2]	−9.1 [−11.2, −6.7]	−10.3 [−12.3, −8.3]
Instruct → AlphaMed, T=0.7	0.28 → 0.32	0.32 → 0.34	0.17 → 0.21
	+4.3 [+2.0, +6.6]	+2.2 [+0.4, +4.0]	+3.9 [+2.4, +5.4]
Instruct → HuatuoGPT-o1, T=0.7	0.28 → 0.36	0.32 → 0.37	0.17 → 0.27
	+8.4 [+6.1, +10.8]	+6.0 [+4.0, +7.9]	+9.5 [+7.5, +11.5]
Instruct → m1, T=0.7	0.28 → 0.41	0.32 → 0.41	0.17 → 0.29
	+13.5 [+10.8, +16.2]	+9.7 [+7.2, +12.2]	+11.4 [+9.2, +13.6]
Base → AlphaMed (base-RL), T=0.7	0.40 → 0.39	0.41 → 0.40	0.28 → 0.27
	−0.7 [−3.0, +1.7]	−0.4 [−2.6, +1.7]	−0.4 [−2.4, +1.7]
Base → `acc` GRPO (8B), T=0.7	0.34 → 0.31	0.44 → 0.42	0.31 → 0.27
	−2.8 [−4.4, −1.3]	−2.1 [−3.7, −0.6]	−4.3 [−6.2, −2.5]
Base → `gm` GRPO (8B), T=0.7	0.34 → 0.32	0.44 → 0.43	0.31 → 0.28
	−1.5 [−3.2, +0.1]	−1.3 [−3.0, +0.3]	−3.0 [−4.8, −1.2]

Table 3: Share of (case, condition) pairs with reference probability ≥ 0.1 in which the condition is a top-1 answer in at least one of 32 samples, by DDXPlus severity. Second rows give differences in points with 95% case-bootstrap intervals. All rows use the 980 test cases; GRPO rows are seed 0. General post-training lowers consideration in every tier, including the can't-miss tier; medical post-training raises it; our GRPO from the base model loses at most 4 points, and with the group reward the severe-tier loss is not significant.

5.6. Asking for reasoning changes the picture. With CoT, the post-trained model's pass@32 rises from 0.607 to 0.835: it leads the base model at every k at T=0.7 (+4.7 [+2.9, +6.6] at k=32) and ties it at T=1 (0.0 [−2.1, +1.9]). Its samples remain far less diverse than the base model's (2.1–2.2 vs. 3.9–6.3 distinct answers), so reasoning recovers coverage at k=32 without restoring sampling diversity. The coverage loss is therefore largest when the model answers directly, as in most clinical decision-support deployments, and reasoning at inference time mitigates it. This matches the medical models: the reasoning-trained descendants widen the differential (§4.10).

Output format. Base models more often give invalid answers: unparsable, or with a top answer outside the 49 conditions. This affects 12% of Qwen3-8B-Base samples at T=0.7 and 25% at T=1 (Qwen2.5-7B: 1% and 21%), against at most 0.3% for the post-trained models. Invalid samples count as wrong, which deflates the base model at large k but inflates the post-trained model's gain at small k. Among valid samples only, the post-trained models still gain +12.0 [+9.8, +14.1] (Qwen3, T=0.7), +18.7 [+16.6, +20.7] (Qwen3, T=1), +12.2 [+10.4, +14.0] and +16.8 [+15.0, +18.5] points (Qwen2.5) at k=1, so a quarter to a third of the raw gain at T=1 is format. On the cases where every sample of both models is valid, the large-k loss persists: −15.2 [−21.9, −8.6] at k=32 for Qwen3 (105 cases) and −10.2 [−12.7, −7.6] for Qwen2.5 (748 cases), both at T=0.7. (At T=1 almost no case has 32 valid base samples.) The reported large-k losses are thus lower bounds.

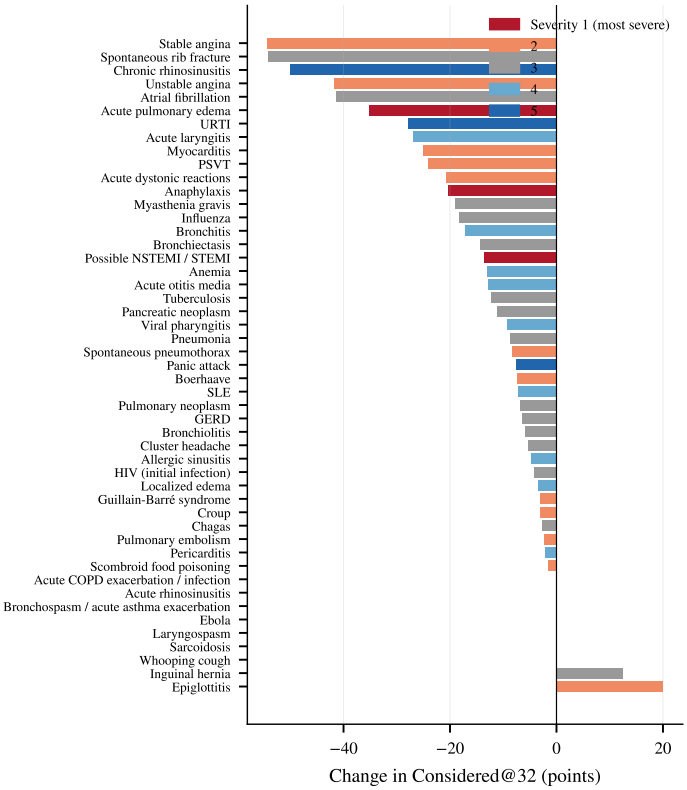


Figure 2: Change in Considered@32 per condition, Qwen2.5-7B-Instruct minus its base model (T=1, 980 cases), colored by DDXPlus severity (dark red = most severe). Conditions with fewer than five cases giving them reference probability ≥ 0.1 are omitted. Angina, acute pulmonary edema, anaphylaxis and myocarditis are among the largest drops.

4.4 Which post-training stage narrows the differential?

The Qwen pairs compare a base model with the end of a post-training pipeline. OLMo 3 [21] releases every stage of its 7B instruct pipeline from one base model: supervised fine-tuning, then DPO, then RL with verifiable rewards on math, code, instruction-following and chat prompts (the released OLMo-3-7B-Instruct). It also releases RL-Zero checkpoints trained with RL directly from the base; we use the general one, trained on chat prompts with an LM-judge reward. (The RL-Zero-Mix checkpoint does not load in our vLLM version.) Table 5 gives the results.

The coverage loss replicates in this third family. At T=0.7, every post-trained stage is worse than the base at k=32: by -14.6 $[-17.5, -11.7]$ points after SFT, -9.2 $[-12.1, -6.1]$ after DPO, -28.3 $[-31.5, -25.0]$ after RLVR, and -13.3 $[-16.1, -10.5]$ for RL-Zero. Diversity falls at every stage of the pipeline, from 4.93 distinct answers for the base to 4.01 after SFT, 3.30 after DPO and 2.37 after RLVR.

Compared stage to stage, DPO recovers part of what SFT lost at large k ($+5.4$ $[+2.9, +8.0]$ points at k=32, and $+10.3$ $[+9.0, +11.6]$ at k=1) while narrowing the differential further (-0.71 distinct answers). The final RLVR stage is the largest single narrowing. Against the DPO model it loses -19.1 $[-21.8, -16.3]$ points at k=32, 0.92 distinct answers and 7.7 points of coverage@32, and it is also worse at k=1 (-5.9 $[-7.2, -4.6]$). It drops severe-tier Considered@32 to 0.25, a loss of -16.1 $[-18.6, -13.5]$ points relative to the base, against -7.4 for the DPO model. At T=1 the pattern is the same (RLVR vs. DPO: -12.7 $[-15.4, -9.9]$ at k=32).

		pass@1		pass@32		Distinct	
Prompt	T	Base	Post	Base	Post	Base	Post
default	0.7	0.388	**0.559**	**0.730**	0.607	3.76	1.27
default	1.0	0.282	**0.556**	**0.789**	0.643	6.60	1.53
plain	0.7	0.352	**0.583**	**0.727**	0.655	3.72	1.28
plain	1.0	0.244	**0.586**	**0.792**	0.705	5.57	1.54
CoT	0.7	0.409	**0.652**	0.788	**0.835**	3.89	2.08
CoT	1.0	0.275	**0.642**	0.837	0.837	6.31	2.24

Table 4: Qwen3-8B-Base vs. Qwen3-8B on DDXPlus (980 cases, 32 samples) under three prompts. Bold marks the significantly better model (paired case bootstrap). Without reasoning, the base model wins at k=32; with CoT, the post-trained model matches or beats it, but its samples stay far less diverse.

RL applied directly to the base narrows the differential less. At T=1, RL-Zero keeps the base model's pass@32 (-1.8 $[-4.5, +0.8]$) and 6.28 of its 7.53 distinct answers, and it loses 4.0 $[1.2, 6.9]$ points of severe-tier consideration, against 14.4 for the full instruct pipeline. At T=0.7 it still loses 13.3 points at k=32, about as much as SFT.

The OLMo base model often breaks the output format: 40% of its samples are invalid at T=0.7 and 47% at T=1, while unparsable completions are 16–21% for the SFT model and at most 1.6% for the later stages. No case has 32 valid base samples, so the all-valid check of §4.3 is not possible here. Among valid samples, the post-trained models' gains at k=1 are small (-1.9 to $+4.0$ points at T=0.7, $+1.3$ to $+8.1$ at T=1), so most of OLMo's raw gain at k=1 is format. Invalid samples count as misses and are excluded from distinct answers and entropy, so the base model's coverage and diversity are understated and the losses above are conservative.

Across three families, then, the collapse of the sampled differential is not a necessary effect of RL. It accumulates over the instruction-tuning pipeline; in OLMo 3 7B its largest step is RL applied on top of SFT and DPO, but at 32B it is SFT (§4.5), so which stage narrows most depends on the recipe. RL applied directly to a base model (OLMo RL-Zero, AlphaMed's base-initialized checkpoint, and our GRPO runs in §5) narrows the differential less. On Phenopacket Store the instruct pipeline also loses coverage: every stage trails the base model at k=32 (-2.0 $[-3.0, -1.1]$ points after SFT, -1.5 after DPO, -1.4 after RLVR; all significant), while RL-Zero does not (-0.6 $[-1.4, +0.2]$), from a base model at 4.5% (§4.11).

4.5 Replication at 32B

We repeat the DDXPlus comparisons at T=0.7 with Qwen2.5-32B and its Instruct model, and with every released stage of OLMo 3.1 32B, whose base is OLMo-3-1125-32B; there is no 32B RL-Zero checkpoint (Table 6). The sampled differential collapses at 32B too, by about half as much. Post-training costs 4.9 $[2.8, 7.1]$ points at pass@32 for Qwen2.5-32B and 4.0 $[2.0, 5.9]$ for OLMo 3.1 32B, halves the number of distinct answers (3.78 $\rightarrow$ 1.68 and 4.32 $\rightarrow$ 2.15), and lowers severe-tier consideration by 6.1 and 8.4 points. For Qwen the loss holds on the 421 cases where every base sample is valid (-4.8 $[-7.8, -1.7]$); the OLMo 32B base gives invalid answers on 27% of samples. The stage attribution differs from 7B. In OLMo 3.1 32B the largest narrowing is SFT (pass@32 -12.1 $[-14.4, -9.9]$, distinct answers -1.39, severe tier -11.9); DPO recovers most of it ($+10.9$ against SFT), and the final RL step is small (-2.8 $[-4.3, -1.1]$ at k=32, -0.14 distinct answers). OLMo 3.1 32B and OLMo 3 7B were post-trained with different recipes, so the stage that narrows the samples most is a property of the pipeline, not of SFT, DPO or RL as such. The stated readout

T	Stage	pass@1	pass@32	Distinct	Entropy	Cov@32	Severe C@32
0.7	Base	0.178	**0.705**	**4.93**	**1.16**	**0.351**	**0.412**
	SFT	0.222	0.559	4.01	0.87	0.257	0.373
	SFT+DPO	**0.324**	0.613	3.30	0.63	0.272	0.338
	SFT+DPO+RLVR (Instruct)	0.266	0.422	2.37	0.42	0.195	0.251
	RL-Zero (RL from base)	0.295	0.572	4.06	0.84	0.286	0.326
1.0	Base	0.128	**0.713**	**7.53**	**1.66**	**0.398**	**0.469**
	SFT	0.191	0.634	6.32	1.29	0.321	0.426
	SFT+DPO	**0.313**	0.688	4.48	0.87	0.312	0.410
	SFT+DPO+RLVR (Instruct)	0.272	0.561	3.58	0.68	0.259	0.325
	RL-Zero (RL from base)	0.279	0.695	6.28	1.21	0.367	0.429

Table 5: OLMo 3 7B post-training stages on DDXPlus (980 cases, 32 samples). Severe C@32 is severe-tier Considered@32 as in Table 3. Bold marks the best value in each block. Diversity falls at every stage of the instruct pipeline, and the RLVR stage loses the most coverage; RL from the base keeps the base model's pass@32 at T=1.

Model	pass@1	pass@32	Distinct	Cov@32	Severe C@32	TV one list↓	TV samples↓	Vote@32 gain
Qwen2.5-32B (base)	0.500	**0.795**	**3.78**	**0.352**	**0.35**	0.726	**0.726**	+7.0
Qwen2.5-32B-Instruct	**0.604**	0.746	1.68	0.276	0.29	**0.674**	0.759	+1.9
OLMo-3-1125-32B (base)	0.326	**0.796**	**4.32**	**0.347**	**0.39**	0.746	**0.742**	+8.4
SFT	0.461	0.674	2.94	0.265	0.27	0.736	0.776	+3.6
SFT+DPO	**0.584**	0.784	2.29	0.293	0.32	**0.694**	0.755	+2.1
SFT+DPO+RLVR (Instruct)	0.560	0.756	2.15	0.283	0.30	0.701	0.763	+0.9

Table 6: 32B replication on DDXPlus (T=0.7, 980 cases, 32 samples). TV columns: distance to the reference differential of one stated list and of the sampled top-1 answers (§4.6). Vote@32 gain: accuracy of the majority answer over 32 samples minus single-call accuracy, in points (§4.7). Bold: best value per family.

improves in both 32B families (§4.6), and voting stops paying off: majority voting over 32 samples gains the base models 7.0 and 8.4 points and the Instruct models 1.9 and 0.9 (n.s.).

4.6 The stated differential does not collapse

Every completion also *states* a differential: up to five conditions with probabilities. We read it directly. For each valid sample we renormalize its stated probabilities over the conditions it names (uniform if the probabilities are missing or invalid) and compare this single list with the reference differential: its total-variation distance to the reference, the reference mass of the conditions it names, and, for each severe condition with reference probability at least 0.1, whether the list names it. For the sampled readout we use the empirical distribution of top-1 answers over the valid samples and, for the same budget of five names, the reference mass covered by the distinct top answers of five random samples. Two oracles bound what each format can achieve: the reference's own top five stated as a list, and as many draws from the reference as the model has valid samples (Table 7, Figure 3).

For the Qwen models the two readouts move in opposite directions under post-training. In both families and at both temperatures, the post-trained model's single list is closer to the reference than the base model's (TV -0.043 to -0.086, all intervals exclude zero), covers more of its mass (+6.1 to +10.8 points), and names severe conditions more often (+9.1 to +13.4 points). Its sampled

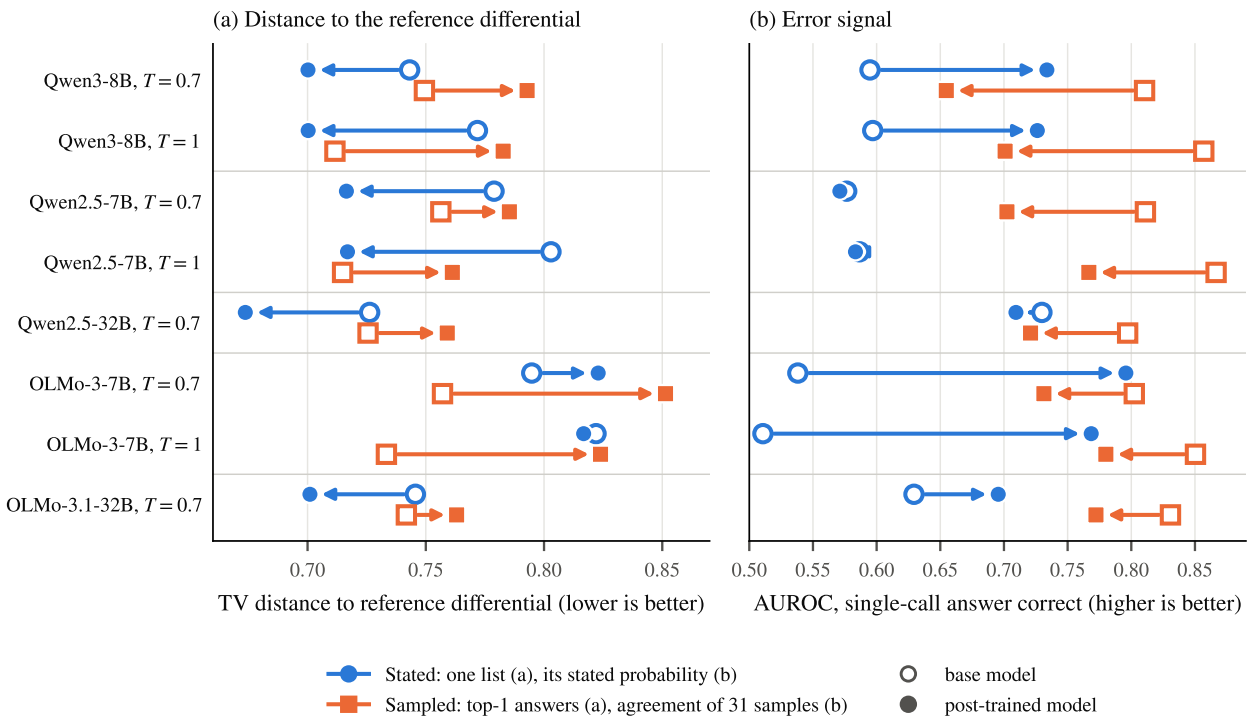


Figure 3: Base (hollow) → post-trained (filled) on DDXPlus (980 cases; OLMo: base → final Instruct; 32B models at T=0.7 only). (a) TV distance to the reference differential of one stated list (blue circles) and of the sampled top-1 answers (orange squares). (b) AUROC for the single-call answer being correct, of its stated probability (blue) and of agreement across 31 further samples (orange).

top answers move farther from the reference (+0.029 to +0.071), and five of them cover less (−1.9 to −6.0 points). Within each post-trained model the list beats its own samples by 0.04–0.09 in TV; within each base model the gap is about zero or favors the samples. The CoT prompt keeps the list's advantage (TV −0.042, severe naming +8.1 points) and removes the loss in the samples (+0.002, n.s.). This quantifies the observation of §4.2 that conditions dropping out of the top answer remain lower in the list.

At 32B the pattern holds in both families (Table 6): the post-trained lists are closer to the reference (TV −0.053 for Qwen2.5-32B-Instruct, −0.052 and −0.045 for the OLMo 3.1 DPO and Instruct models) and cover 5.6–6.7 points more of it, while the samples move away (+0.014 to +0.034). Within every post-trained 32B model the list beats its own samples by 0.04–0.09 in TV, against about zero for both base models. OLMo 3 7B is the exception with the five-item prompt. Its DPO and Instruct models' lists are no closer to the reference than the base model's (TV +0.000 and +0.028 at T=0.7; −0.026 and −0.005 at T=1) and cover at most 5 points more of it; only the naming of severe conditions improves (+6.1 to +12.8 points). Asked for a probability on every condition, its Instruct model's stated distribution is closer to the reference than the base model's (−0.032 [−0.038, −0.027]; §4.8), so the exception is specific to the short list. RL-Zero's lists improve slightly (TV −0.020 and −0.039). The medical descendants change little relative to their parent (TV −0.012 for AlphaMed, −0.014 for HuatuoGPT-o1, +0.017 for m1). In absolute terms every model is far from the reference: a single list covers 0.22–0.41 of the reference mass, against 0.74 for the reference's own top five.

Pair	Model	TV to reference↓ One list	TV to reference↓ Samples	Reference mass covered One list	Reference mass covered 5 samples	Severe named One list
Qwen3-8B, T=0.7	base	0.743	**0.749**	0.335	**0.245**	0.347
	post-trained	**0.700**	0.793	**0.396**	0.207	**0.437**
Qwen3-8B, T=1	base	0.772	**0.712**	0.293	**0.275**	0.312
	post-trained	**0.700**	0.783	**0.396**	0.215	**0.440**
Qwen2.5-7B, T=0.7	base	0.779	**0.756**	0.291	**0.233**	0.345
	instruct	**0.716**	0.785	**0.364**	0.214	**0.440**
Qwen2.5-7B, T=1	base	0.803	**0.715**	0.255	**0.257**	0.304
	instruct	**0.717**	0.761	**0.363**	0.234	**0.438**
OLMo 3 7B, T=0.7	base	0.795	**0.757**	0.268	**0.237**	0.256
	SFT+DPO	0.795	0.804	0.277	0.193	**0.378**
	Instruct (RLVR)	0.823	0.851	0.257	0.150	0.317
	RL-Zero	**0.775**	0.793	**0.295**	0.197	0.274
Oracle	reference	0.265	0.18–0.25	0.735	0.563	0.975

Table 7: Stated vs. sampled differentials on DDXPlus (980 cases, valid samples only). *One list*: a single sample's stated differential, renormalized over the conditions it names (mean over samples). *Samples*: the empirical distribution of top-1 answers over the 32 samples; *5 samples*: reference mass of the distinct top answers in five random samples, the same five-name budget as one list. *Severe named*: share of (case, severe condition with reference probability ≥ 0.1) pairs that one list names. Bold marks the better model of each pair (OLMo: of the block). Oracle: the reference's own top five as a list, and draws from the reference as samples. Post-training improves the Qwen lists while worsening their samples; OLMo's lists do not improve.

4.7 Consequences for sampling-based methods

We ask what the gap does to two common uses of repeated samples: aggregating them into an answer, and using their agreement as an uncertainty signal (Table 8). The single-call answer is the top answer of a model's first valid sample.

Aggregation. Majority voting over 32 samples (self-consistency; 23) raises base models' accuracy by 3.4–5.2 points at T=0.7 and 10.4–14.2 at T=1. For the general post-trained models it adds at most 0.8 points at T=0.7 (Qwen3-8B +0.3, Qwen2.5-7B-Instruct +0.8, OLMo-3-Instruct −0.2; all n.s.) and 0.1–4.4 at T=1. Models that reason before answering still benefit (HuatuoGPT-o1 +4.8, m1 +3.8, Qwen3-8B with the CoT prompt +4.5). Averaging the stated lists instead of voting changes accuracy by at most about 2 points for any model. For a top-three differential, the stated list is the better source after post-training. For Qwen3-8B, the top three of one list contains the true pathology 13.7 [+11.4, +15.9] points more often than the three most frequent answers in 32 samples (T=1: +10.0), and for Qwen2.5-7B-Instruct 7.3 [+5.3, +9.6] points more often (T=1: +2.0, n.s.). For every base model the 32 samples win, by 7.3–13.6 points at T=0.7. OLMo-3-Instruct is in between (+1.9, n.s.; T=1: −4.7).

Uncertainty. We score how well each signal ranks the correctness of the single-call answer (AUROC). The stated probability of that answer needs no extra calls; *agreement*, the share of the model's other samples with the same top answer, uses 4 or 31 more. For base models agreement is far better: 0.80–0.81 with 31 extra samples, against 0.54–0.60 for their nearly uninformative stated probability. Post-training weakens agreement in every family (Qwen3 0.810 → 0.655, Qwen2.5

Model	Single-call accuracy	Vote@32 gain	Top-3: list − 32 samples	AUROC, single-call answer correct		
				Stated p	Agree@5	Agree@32
Qwen3-8B-Base	0.456	+3.4	−7.3	0.595	0.772	**0.810**
Qwen3-8B	0.556	+0.3	+13.7	**0.734**	0.613	0.655
Qwen3-8B (CoT prompt)	0.648	+4.5	+0.0	0.718	0.725	**0.778**
Qwen2.5-7B	0.371	+5.2	−8.9	0.576	0.764	**0.811**
Qwen2.5-7B-Instruct	0.480	+0.8	+7.3	0.571	0.653	**0.702**
OLMo-3 base	0.291	+4.0	−13.6	0.538	0.761	**0.802**
OLMo-3 SFT+DPO	0.327	−0.4	−7.8	0.738	0.702	**0.751**
OLMo-3-Instruct	0.267	−0.2	+1.9	**0.796**	0.668	0.731
AlphaMed-7B	0.528	+0.9	+0.6	0.673	0.663	**0.715**
HuatuoGPT-o1-7B	0.532	+4.8	−0.4	0.590	0.715	**0.775**
m1-7B	0.495	+3.8	−4.6	0.680	0.736	**0.782**

Table 8: Aggregation and uncertainty on DDXPlus (T=0.7, 980 cases). Single-call answer: top-1 of the first valid sample. Vote@32 gain: accuracy of the majority top-1 over all valid samples minus single-call accuracy, in points. Top-3: rate at which one stated list's top three contains the true pathology, minus the rate for the three most frequent sampled answers (points). AUROC for the single-call answer being correct: its stated probability (no extra calls) vs. agreement of 4 or 31 further samples; bold marks the best signal. Post-trained models gain little from voting; for some of them the stated probability becomes the better error signal.

0.811 → 0.702, OLMo 0.802 → 0.731), consistent with Liu [16], and moves the comparison toward the stated probability. For Qwen3-8B (0.734) and OLMo-3-Instruct (0.796), the stated probability is better than agreement over 32 samples (+0.079 [+0.046, +0.110] and +0.064 [+0.027, +0.101]); at T=1 the two are statistically tied. For Qwen2.5-7B-Instruct (0.571 vs. 0.702) and all three medical models, agreement remains better. These AUROCs are computed on each model's own answers, so they compare signals within a model, not models with each other.

Ambiguity. The reference differential also says how ambiguous each case is. We correlate each uncertainty signal with the entropy of the case's reference differential. For every model and signal the Spearman correlation lies between −0.10 and +0.14, except AlphaMed's stated probability and stated entropy (+0.30 and +0.32). How uncertain a model is about a case, by either readout, says almost nothing about how ambiguous the case is.

At 32B. The same shifts appear with smaller margins. Voting gains fall from 7.0 and 8.4 points for the base models to 1.9 and 0.9 (n.s.) for the Instruct models, and agreement weakens as an error signal (Qwen 0.797 → 0.721, OLMo 0.831 → 0.772). For Qwen2.5-32B-Instruct the top three of one list beats the three most frequent of 32 sampled answers by 5.6 [3.6, 7.7] points, but its stated probability only ties agreement over 32 samples (−0.011, n.s.; it beats agreement over five, +0.047); for OLMo 3.1 32B Instruct agreement stays better (−0.077).

After post-training, then, repeated sampling adds little: voting no longer improves accuracy and agreement carries less signal, while one stated list carries as much of the differential or more. Where the crossover falls depends on the model, so neither readout can be assumed to be the informative one without checking.

Model	TV to reference↓ One distribution	Samples	AUROC, single-call answer correct Stated p	Agree@5	Agree@32	Top-3: one dist. − 32 samples
Qwen3-8B-Base	0.733	**0.752**	0.647	0.809	**0.845**	−6.7
Qwen3-8B	**0.681**	0.790	**0.717**	0.609	0.647	+14.6
Qwen2.5-7B	0.769	**0.758**	0.688	0.792	**0.827**	−10.2
Qwen2.5-7B-Instruct	**0.702**	0.770	0.674	0.715	**0.748**	+1.6
OLMo-3-1025-7B	0.807	**0.766**	0.610	0.744	**0.786**	−15.3
OLMo-3-7B-Instruct	**0.775**	0.787	0.749	0.705	**0.759**	−9.5

Table 9: Full-distribution prompt on DDXPlus (T=0.7, 980 cases): a probability on every condition, top answer = highest stated probability. TV: one sample's stated distribution vs. the sampled top-1 answers (bold: better model of each pair). AUROC: bold marks each model's best signal.

4.8 Robustness of the stated readout to elicitation

The stated readout above uses a five-item list. We repeat the Qwen pairs and OLMo 3 7B's base and Instruct models at T=0.7 with a prompt that asks for a probability on every one of the 49 conditions, summing to one, and take each sample's top answer to be its highest-probability condition (Table 9). The results match the short list. After post-training the stated distribution is closer to the reference (TV −0.052 for Qwen3-8B, −0.067 for Qwen2.5-7B-Instruct, −0.032 for OLMo-3-7B-Instruct) while the samples move away (+0.012 to +0.038). Qwen3-8B's stated probability again beats agreement over 32 samples (+0.069 [+0.033, +0.104]), Qwen2.5-7B-Instruct's again does not (−0.074), and OLMo-3-7B-Instruct's ties it (−0.010; +0.044 against five samples). The top three of one full distribution beats the three most frequent of 32 sampled answers for Qwen3-8B (+14.6 points) but not for the other two models.

4.9 Which readout to trust: a cheap audit

Whether the stated probability or agreement across samples is the better error signal varies across models, so we ask what predicts it and whether a practitioner can tell cheaply. We pool all 56 model–setting combinations we scored on DDXPlus: the base/post-trained pairs at both temperatures and with three prompts, the OLMo stages, the 32B models, the medical models, our GRPO runs, and the full-distribution prompt. Across them, the stated probability's advantage over five-sample agreement correlates with the *single-answer rate*, the share of cases whose five samples all give the same top answer (Spearman 0.75), and with the stated probability's own AUROC (0.82); it correlates negatively with the number of distinct answers (−0.74). The stated probability is the better signal for 19 of the 56. No single-answer threshold separates them: Qwen2.5-7B-Instruct still favors agreement, while OLMo 3 7B's DPO and Instruct models favor their unusually informative stated probability. A small audit settles it for a given model. Using only five samples per case on 100 random cases, choose the signal with the higher AUROC. Across 500 draws per model, this choice matches the one made on all 980 cases in 88% of draws on average, and when it does not, the two signals are close, so the mean AUROC lost is 0.004 (50 cases: 83%, 0.009; 200 cases: 92%, 0.002). The worst model is OLMo 3.1 32B DPO (43%), whose two signals tie (0.693 vs. 0.694).

4.10 Medical post-training

Applied after instruction tuning, medical RL does not cost coverage. AlphaMed improves on its parent at every k (Figure 4): +3.9 points at k=1 ([+2.2, +5.5]), widening to +8.4 at k=32 ([+6.0, +10.6]). It reaches at least one correct sample for 47 of 49 conditions, against 39 for its

Model	Post-training	pass@1	pass@32	Distinct	Entropy	Cov@32	ECE
Qwen2.5-7B-Instruct	(parent)	0.478	0.598	1.82	0.30	0.239	**0.106**
AlphaMed-7B	rule-based RL	0.516	0.682	1.87	0.28	0.266	0.121
HuatuoGPT-o1-7B	SFT + PPO	**0.536**	0.735	2.49	0.44	0.306	0.168
m1-7B-23K	SFT distillation	0.486	0.779	3.29	0.62	0.328	0.206
Qwen2.5-7B (base)	(parent)	0.352	0.695	**4.91**	**1.00**	**0.335**	0.291
AlphaMed-7B (base-RL)	rule-based RL	0.495	**0.790**	3.14	0.62	0.320	0.202

Table 10: Medical models on DDXPlus (980 cases, 32 samples, T=0.7). Upper block: descendants of Qwen2.5-7B-Instruct; none narrows the differential further, and the reasoning-trained models widen it. Lower block: AlphaMed's companion checkpoint trained from the base model, which keeps most of the base model's diversity and has the best pass@32 of any 7B model.

parent. It does not restore breadth, though: distinct answers (1.87 vs. 1.82), entropy (0.29 vs. 0.30), and coverage (0.27 vs. 0.24) stay at the instruction-tuned level, far below the base model. AlphaMed also names off-list conditions more often than its parent, a mild loss of instruction following. By condition, AlphaMed slightly *raises* consideration in every severity tier (+2.2 to +4.3 points; Table 3). It still loses acute pulmonary edema (0.31 to 0.13), and it introduces an attractor of its own: spontaneous pneumothorax as the top answer in 10.3% of samples, against 4.3% for its parent and a reference rate of 1.0%. Medical post-training thus operates on an already collapsed distribution; the collapse comes from the general post-training stage. The other two descendants go further (Table 10). HuatuoGPT-o1 (SFT then PPO) and m1 (SFT distillation from a reasoning teacher) both *widen* the differential relative to their parent: 2.49 and 3.29 distinct answers vs. 1.82, and coverage@32 of 0.31 and 0.33 vs. 0.24. Their pass@32 rises by 13.7 $[+11.3, +16.0]$ and 18.1 $[+15.5, +20.7]$ points. m1 gains almost nothing at k=1 (+0.9, interval includes zero), so its improvement lies entirely in breadth. Both reasoning-trained models are also less calibrated (ECE 0.17 and 0.21 vs. 0.11). Medical post-training on an instruction-tuned model thus never adds to the collapse, and reasoning distillation partially reverses it.

Medical RL from the base model. Alongside the instruct-initialized AlphaMed-7B, the AlphaMed authors released a companion checkpoint, AlphaMed-7B-base-rl, initialized from Qwen2.5-7B rather than Qwen2.5-7B-Instruct (according to its configuration file). The paper [15] describes only the instruct-initialized models: GRPO with a binary correctness reward on about 19k multiple-choice questions from MedQA and MedMCQA, with a 64-question by 8-sample batch for 300 steps. It does not document the base-initialized checkpoint, so we cannot rule out differences in data or hyperparameters, and we treat it as corroborating rather than controlled evidence. It offers a third-party check on whether RL itself causes the collapse (Table 10, lower block). It does not. RL from the base model improves on the base at every k (+14.3 $[+13.1, +15.7]$ points at k=1, +9.5 $[+7.4, +11.6]$ at k=32). It keeps 3.14 distinct answers, against 1.87 for the instruct-initialized AlphaMed. Against that instruct-model AlphaMed, it is 2.2 $[0.4, 3.9]$ points worse at k=1 and 10.8 $[+8.6, +13.2]$ points better at k=32. It also preserves can't-miss consideration: severe-tier Considered@32 changes by -0.7 $[-3.0, +1.7]$ points relative to the base, against -12.4 for instruction tuning and -8.1 for instruction tuning followed by AlphaMed's RL (Table 3). This is consistent with our own controlled GRPO-from-base result (§5): with a verifiable correctness reward, RL from a base model buys accuracy without the collapse, which is introduced by general instruction tuning and inherited by everything trained on top of it. OLMo 3 refines this (§4.4): within its instruct pipeline, the RL stage applied after SFT and DPO is the largest single narrowing.

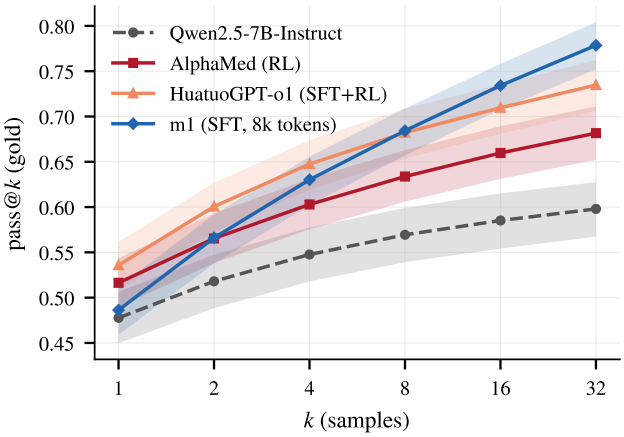


Figure 4: DDXPlus at T=0.7 (980 cases): Qwen2.5-7B-Instruct (dashed) vs. its medical descendants, with 95% case-bootstrap bands. Every descendant matches or beats its parent at every k, and the gaps widen with k: there is no sharpening tax at the medical stage.

4.11 Rare disease: overconfidence without accuracy

On 1,401 Phenopacket Store cases covering 749 diseases, every 7–8B model is near the floor: pass@1 is 1.4–2.2% and pass@32 is 3.7–7.0%. Stated confidence is not near the floor. Qwen2.5-7B-Instruct states a mean top-1 probability of 0.51, and its RL-trained medical descendants state 0.67 (HuatuoGPT-o1) and 0.69 (AlphaMed), while their accuracy is about 2%. ECE accordingly rises from 0.49 to 0.64–0.67. Medical RL thus increases overconfidence exactly where the model is out of its depth. The effect also appears without instruction tuning: AlphaMed's base-initialized checkpoint is the most overconfident model we evaluate (mean confidence 0.71, ECE 0.70, accuracy 1.7%). The medical descendants do gain a little accuracy: AlphaMed beats its parent at every k (+0.6 [+0.2, +0.9] points at k=1, +1.0 [+0.1, +2.1] at k=32), HuatuoGPT-o1 at $k \leq 8$, and m1 at $k \geq 2$ (+1.4 [+0.4, +2.5] at k=32). These gains are an order of magnitude smaller than the rise in confidence. m1, trained only with supervised reasoning traces (at its intended 8k-token budget), is the exception on calibration and diversity: it is the most diverse model (15.3 distinct answers per 32 samples, against 11.5 for the parent and 9.2 for HuatuoGPT-o1), and its confidence rises least (0.57).

General post-training reproduces the DDXPlus pattern on Qwen3. Qwen3-8B beats its base at pass@1 (+0.7 [+0.4, +1.1]) and trails it from k=8 on, by −2.2 [−3.1, −1.3] points at pass@32 (a relative loss of 37%), with half the distinct answers (6.4 vs. 13.4). For Qwen2.5 no pass@k difference is significant. Unlike medical RL, general post-training *lowers* stated confidence here (0.61 to 0.46 for Qwen3, 0.61 to 0.51 for Qwen2.5), although base models also give invalid answers more often (8.5% of Qwen3-8B-Base samples). OLMo 3 does not share this: every stage of its pipeline states a mean confidence of 0.73–0.84 at about 1% accuracy (pass@32 2.5–4.5%), and its SFT checkpoint is the most overconfident model we evaluate (confidence 0.84, ECE 0.83). On the 500-case subset, our 8B GRPO runs (§5) are indistinguishable from their base (all pass@k differences within ±0.6 points).

Accuracy depends strongly on when the source case report was published, which suggests memorization (Table 11). For every model, pass@32 is 5–10% on cases published up to 2022 (n=901), at most 1.7% on 2023–24 cases (n=348), and 2.6–4.6% on 2025+ cases (n=152). Even the low absolute accuracy is therefore inflated by older, likely-memorized cases.

Model	≤2022 (n=901) pass@1	pass@32	2023–24 (n=348) pass@1	pass@32	2025+ (n=152) pass@1	pass@32
Qwen2.5-7B (base)	2.0	8.2	0.0	0.3	0.6	3.3
Qwen2.5-7B-Instruct	2.3	7.7	0.0	0.9	0.8	3.9
AlphaMed-7B	3.0	9.0	0.3	1.4	1.4	3.9
AlphaMed-7B (base-RL)	2.4	8.8	0.1	1.1	1.4	4.6
HuatuoGPT-o1-7B	3.1	8.4	0.3	1.4	1.3	3.9
m1-7B	2.6	9.5	0.3	1.7	1.0	3.9
Qwen3-8B-Base	2.1	8.3	0.1	0.6	1.1	3.9
Qwen3-8B	3.1	5.2	0.1	0.3	2.0	2.6

Table 11: Phenopacket Store accuracy (%, exact diagnosis) by publication year of the source case report (1,401 cases, 32 samples, T=0.7). Every model is far more accurate on cases published up to 2022 than on 2023–24 cases, consistent with memorization of older case reports.

5 Training without collapsing the samples

The stated list survives post-training, but sampled diversity still matters: self-consistency and other sampling-based methods depend on it (§4.7), and so does reinforcement learning itself, since GRPO learns nothing from a group whose samples agree. Can reinforcement learning raise diagnostic accuracy without collapsing the sampled differential? We fine-tune Qwen3-1.7B with GRPO on DDXPlus training patients (9,800 cases, 200 per condition; disjoint from the test split) and evaluate exactly as in §3.

Setup. We train LoRA adapters (r=32) with GRPO for 200 steps. Each step uses 16 prompts and 16 sampled completions per prompt, generated with vLLM on the same GPU. A first sweep with a common aggressive configuration (learning rate 2×10^{-5}, no KL penalty, 8 samples per prompt, per-group reward standardization) was unstable. In three of seven runs, the policy collapsed to an input-independent answer: one condition for 88–100% of all patients, with completions shrinking to about 25 tokens. The starting policy is nearly deterministic (token entropy ≈0.06), so most GRPO groups have zero reward variance, and noisy updates drive the collapse. We therefore use learning rate 5×10^{-6}, KL coefficient 0.04, no reward standardization (as in Puri et al. [20]), and 16 samples per prompt. A monitor tracks the number of distinct top-1 answers per case within each batch. All runs in this configuration were stable.

Rewards. Every recipe includes a format reward: the answer must parse into a list of distinct conditions with valid probabilities. `acc` is standard RLVR: 1 if the top-1 answer is the true pathology. `dist` adds two per-sample distributional terms to `acc`: a Brier score of the stated distribution against the truth, and the overlap of the stated distribution with the reference differential. Both are computed from a *single* completion. They cannot see that the policy gives the same answer in every sample, which is exactly the collapse of §4.2. `gm` adds to `acc` a *group-level* reward over the GRPO group of n samples for a case. If sample i's top-1 answer is a, and k_a samples in the group give a,

$$r_i^{\mathrm{gm}} = \mathrm{clip}\big(n\,p_{\mathrm{ref}}(a) - (k_a - 1),\ 0,\ 1\big), \tag{1}$$

where p_{ref} is the reference differential. This is sample i's marginal contribution to $n\sum_c \min(q_c, p_{\mathrm{ref}}(c))$, the overlap between the group's empirical answer distribution q and the reference. An answer earns credit while its condition is under-represented relative to its reference mass, and none once it is

over-represented. In contrast to multi-answer rewards that score a set inside one completion [20], `gm` acts on the distribution of *sampled* answers, the quantity that collapses. Because the group reward fires only when a group contains dissenting samples, we also test a boosted variant, `gm`$^+$: `gm` weight 3 and training temperature 1.2. As baselines we adapt the multi-answer rewards of Puri et al. [20] to our prompt. MULTI-RLVR scores the share of listed conditions that belong to the reference differential, treated as a set of acceptable answers; MULTI-RLCR adds their set-level Brier term on the per-answer probabilities. Both zero the reward for duplicate or invalid answers, as in the original, and use the same format reward, optimizer and settings as our recipes.

Results. Table 13 reports means over two seeds. All recipes raise top-1 accuracy by 8–10 points. The per-sample recipes (`acc`, `dist`) slightly *reduce* diversity relative to the starting model, the sharpening documented in §4.1. The group reward removes this. `gm` matches the starting model's diversity, and `gm`$^+$ exceeds it while also being the most accurate recipe. Against `acc`, with paired bootstraps over cases averaged over seeds, `gm`$^+$ improves:

- pass@1 by $+1.6$ $[+1.0, +2.3]$ and pass@32 by $+1.7$ $[+1.0, +2.6]$ points;
- distinct answers by $+0.10$ $[+0.07, +0.13]$ and entropy by $+0.034$ $[+0.022, +0.047]$;
- coverage@32 by $+1.5$ $[+1.0, +2.1]$ points, and TV by -0.014 $[-0.019, -0.010]$.

Plain `gm` improves on `acc` in the same direction with smaller margins (e.g. pass@1 $+1.0$ $[+0.6, +1.6]$, entropy $+0.013$ $[+0.005, +0.021]$). Diversity increases monotonically with group-reward pressure, with no accuracy cost. The set-membership baselines of Puri et al. [20] train stably (their reward rises from 0.27–0.31 to 0.32–0.35) but barely change sampled behavior. pass@1 rises by under 1 point over the untrained model, and diversity is unchanged. Against them, `gm`$^+$ is better on every metric, by $+8.9$ $[+7.4, +10.5]$ points at pass@1, $+0.07$ $[+0.03, +0.11]$ distinct answers and $+3.0$ $[+2.3, +3.8]$ points of coverage@32 (vs. MULTI-RLVR; MULTI-RLCR is similar). What they do improve is the differential written *inside* one completion: the reference mass covered by a single listed differential rises to 0.317 (MULTI-RLVR), against 0.300 for `acc` and 0.309 for `gm`$^+$. The two objectives are complementary: a set reward shapes the list a model writes, while the group reward shapes which answer it commits to across samples, which is what collapses under post-training. The absolute effects are modest because the starting policy is already collapsed: there is little exploration for any reward to act on. None of the recipes changes rare-disease accuracy or confidence on Phenopacket Store, where a 1.7B model is at the floor; the same holds at 8B (§4.11).

Starting from a diverse base model. The 1.7B experiment starts from an already collapsed instruct model. We therefore repeat the comparison from Qwen3-8B-Base, the diverse base model of §4.1, with the same recipe and `acc` vs. `gm`. We compare against Qwen's own post-trained Qwen3-8B (Table 12; means of two seeds). Both RL recipes reach nearly the official model's top-1 accuracy (0.537 and 0.539 vs. 0.558; `gm`: -1.9 $[-3.9, +0.1]$ points). They keep more than twice its diversity (2.7–2.9 vs. 1.27 distinct answers) and beat it by 15–16 points at pass@32 (`gm`: $+16.3$ $[+14.0, +18.8]$). The collapse produced by general post-training is therefore not the price of its top-1 accuracy. Both runs still narrow the base model somewhat, and the group reward narrows it less. At equal pass@1 ($+0.2$, n.s.), `gm` beats `acc` on distinct answers $+0.14$ $[+0.11, +0.17]$, entropy $+0.035$ $[+0.029, +0.041]$, coverage@32 $+1.1$ $[+0.8, +1.5]$ points, and pass@32 $+1.2$ $[+0.5, +1.9]$. Each seed reproduces the ordering. The severe tier tells the same story (Table 3): official post-training removes 9.8 points of can't-miss consideration, `acc` 2.8, and `gm` 1.5 (not significant). Figure 5 shows the pass@k curves.

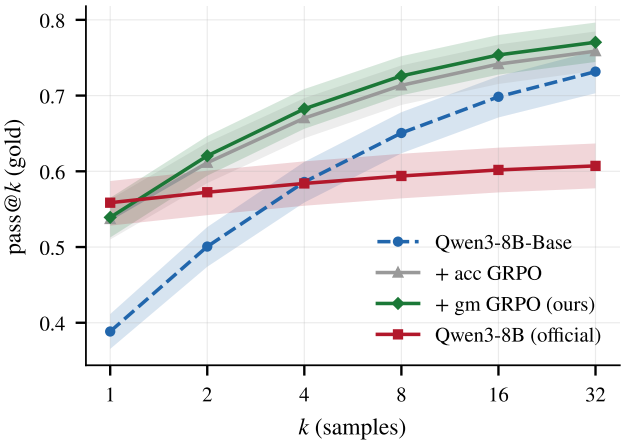


Figure 5: pass@k on DDXPlus (T=0.7, 980 cases) from the same pretrained weights: Qwen3-8B-Base, our GRPO runs (pooled over two seeds), and the official post-trained Qwen3-8B, with 95% case-bootstrap bands. Our RL lifts the whole curve; official post-training trades the right end for the left.

Model	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-8B-Base (start)	0.389	0.732	**3.77**	**0.78**	**0.323**	0.749
+ `acc` GRPO (ours)	0.537	0.759	2.72	0.52	0.297	0.750
+ `gm` GRPO (ours)	0.539	**0.770**	2.86	0.56	0.308	**0.742**
Qwen3-8B (official post-training)	**0.558**	0.607	1.27	0.11	0.217	0.793

Table 12: GRPO from Qwen3-8B-Base vs. the official post-trained Qwen3-8B on DDXPlus (980 cases, 32 samples, T=0.7); GRPO rows are means of two seeds. Our RL nearly matches the official model's top-1 accuracy while keeping more than twice its diversity; the group reward preserves more than `acc`.

6 Limitations

DDXPlus is synthetic: its reference differential is the posterior of a rule-based engine, and the simulated pathology is not always ranked first in it (it is for "more than 70%" of patients [7, §3.5]), so we treat TV and coverage as distance to a *reference* posterior rather than to clinical truth. Base models more often give invalid answers. We count these as errors, which makes the large-k losses conservative but inflates post-trained gains at small k; conditioning on valid samples removes a quarter to a third of the T=1 pass@1 gain (§4.3). Our base/post-trained comparisons cover two open families at 7–32B; prompt robustness of the sampled readout is tested on one pair, and the collapse at large k largely disappears when the model is asked to reason first (§4.3). The OLMo base models give invalid answers on 27–47% of samples, so their format-controlled comparisons are weaker than Qwen's. We evaluate one RL-Zero checkpoint, at 7B, trained with an LM-judge reward on chat prompts rather than a diagnosis reward, and the 32B models only at T=0.7. Phenopacket cases are drawn from published literature and may be in pretraining data, which we partly address with era splits. The stated readout renormalizes each list's probabilities over the conditions it names, and the single-call answer is a model's first valid sample. A full-distribution prompt gives the same picture (§4.8), but other elicitations may differ. Which readout is the better error signal is model-dependent; the audit of §4.9 makes it checkable but does not remove the dependence. We do not evaluate clinical utility with clinicians.

Recipe	pass@1	pass@32	Distinct	Entropy	Cov@32	TV↓
Qwen3-1.7B (untrained)	0.378	0.435	1.41	0.151	0.162	0.850
`acc`	0.458	0.499	1.36	0.142	0.181	0.829
`dist`	0.458	0.502	1.34	0.131	0.181	0.829
MULTI-RLVR [20]	0.385	0.443	1.39	0.153	0.166	0.845
MULTI-RLCR [20]	0.383	0.441	1.39	0.150	0.165	0.847
`gm`	0.469	0.510	1.40	0.155	0.186	0.825
$\texttt{gm}^{+}$	**0.474**	**0.516**	**1.46**	**0.176**	**0.196**	**0.815**

Table 13: GRPO on Qwen3-1.7B, evaluated on DDXPlus test (980 cases, 32 samples, T=0.7); means of two seeds. $\texttt{gm}^{+}$: group-reward weight 3, training temperature 1.2. Per-sample rewards slightly narrow the model; set-membership rewards barely move it; the group reward preserves, then widens, its differential while improving accuracy.

References

[1] Junying Chen, Zhenyang Cai, Ke Ji, Xidong Wang, Wanlong Liu, Rongsheng Wang, and Benyou Wang. Towards medical complex reasoning with LLMs through medical verifiable problems. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 14552–14573, Vienna, Austria, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.findings-acl.751. arXiv version titled "HuatuoGPT-o1, Towards Medical Complex Reasoning with LLMs".

[2] Mark Chen, Jerry Tworek, Heewoo Jun, Qiming Yuan, Henrique Ponde de Oliveira Pinto, Jared Kaplan, Harri Edwards, Yuri Burda, Nicholas Joseph, Greg Brockman, Alex Ray, Raul Puri, Gretchen Krueger, Michael Petrov, Heidy Khlaaf, Girish Sastry, Pamela Mishkin, Brooke Chan, Scott Gray, Nick Ryder, Mikhail Pavlov, Alethea Power, Lukasz Kaiser, Mohammad Bavarian, Clemens Winter, Philippe Tillet, Felipe Petroski Such, Dave Cummings, Matthias Plappert, Fotios Chantzis, Elizabeth Barnes, Ariel Herbert-Voss, William Hebgen Guss, Alex Nichol, Alex Paino, Nikolas Tezak, Jie Tang, Igor Babuschkin, Suchir Balaji, Shantanu Jain, William Saunders, Christopher Hesse, Andrew N. Carr, Jan Leike, Josh Achiam, Vedant Misra, Evan Morikawa, Alec Radford, Matthew Knight, Miles Brundage, Mira Murati, Katie Mayer, Peter Welinder, Bob McGrew, Dario Amodei, Sam McCandlish, Ilya Sutskever, and Wojciech Zaremba. Evaluating large language models trained on code. 2021.

[3] Zhoujun Cheng, Shibo Hao, Tianyang Liu, Fan Zhou, Yutao Xie, Feng Yao, Yuexin Bian, Yonghao Zhuang, Nilabjo Dey, Yuheng Zha, Yi Gu, Kun Zhou, Yuqi Wang, Yuan Li, Richard Fan, Jianshu She, Chengqian Gao, Abulhair Saparov, Haonan Li, Taylor W. Killian, Mikhail Yurochkin, Zhengzhong Liu, Eric P. Xing, and Zhiting Hu. Revisiting reinforcement learning for LLM reasoning from a cross-domain perspective. In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[4] Mehul Damani, Isha Puri, Stewart Slocum, Idan Shenfeld, Leshem Choshen, Yoon Kim, and Jacob Andreas. Beyond binary rewards: Training LMs to reason about their uncertainty. In *International Conference on Learning Representations (ICLR)*, 2026.

[5] Daniel Danis, Michael J. Bamshad, Yasemin Bridges, Andrés Caballero-Oteyza, Pilar Cacheiro, Leigh C. Carmody, Leonardo Chimirri, Jessica X. Chong, Ben Coleman, Raymond Dalgleish, Peter J. Freeman, Adam S. L. Graefe, Tudor Groza, Peter Hansen, Julius O. B. Jacobsen,

Adam Klocperk, Maaike Kusters, Markus S. Ladewig, Allison J. Marcello, Teresa Mattina, Christopher J. Mungall, Monica C. Munoz-Torres, Justin T. Reese, Filip Rehburg, Bárbara C. S. Reis, Catharina Schuetz, Damian Smedley, Timmy Strauss, Jagadish Chandrabose Sundaramurthi, Sylvia Thun, Kyran Wissink, John F. Wagstaff, David Zocche, Melissa A. Haendel, and Peter N. Robinson. A corpus of GA4GH phenopackets: Case-level phenotyping for genomic diagnostics and discovery. *Human Genetics and Genomics Advances*, 6(1):100371, 2025. doi: 10.1016/j.xhgg.2024.100371.

[6] Arsene Fansi Tchango, Rishab Goel, Julien Martel, Zhi Wen, Gaetan Marceau Caron, and Joumana Ghosn. Towards trustworthy automatic diagnosis systems by emulating doctors' reasoning with deep reinforcement learning. In *Advances in Neural Information Processing Systems (NeurIPS 2022)*, volume 35, 2022.

[7] Arsene Fansi Tchango, Rishab Goel, Zhi Wen, Julien Martel, and Joumana Ghosn. DDXPlus: A new dataset for automatic medical diagnosis. In *Advances in Neural Information Processing Systems (NeurIPS 2022), Datasets and Benchmarks Track*, 2022.

[8] Sebastian Farquhar, Jannik Kossen, Lorenz Kuhn, and Yarin Gal. Detecting hallucinations in large language models using semantic entropy. *Nature*, 630(8017):625–630, 2024.

[9] Xiaoke Huang, Juncheng Wu, Hui Liu, Xianfeng Tang, and Yuyin Zhou. m1: Unleash the potential of test-time scaling for medical reasoning with large language models. In *Proceedings of the Fifth Machine Learning for Health Symposium (ML4H 2025)*, volume 297 of *Proceedings of Machine Learning Research*, pages 369–383. PMLR, 2026.

[10] Ahmadreza Jeddi, Kimia Shaban, Negin Baghbanzadeh, Natasha Sharan, Abhishek Moturu, Elham Dolatabadi, and Babak Taati. When does RL help medical VLMs? disentangling vision, SFT, and RL gains. 2026.

[11] Liwei Jiang, Yuanjun Chai, Margaret Li, Mickel Liu, Raymond Fok, Nouha Dziri, Yulia Tsvetkov, Maarten Sap, Alon Albalak, and Yejin Choi. Artificial hivemind: The open-ended homogeneity of language models (and beyond). In *Advances in Neural Information Processing Systems (NeurIPS 2025), Datasets and Benchmarks Track*, 2025.

[12] Robert Kirk, Ishita Mediratta, Christoforos Nalmpantis, Jelena Luketina, Eric Hambro, Edward Grefenstette, and Roberta Raileanu. Understanding the effects of RLHF on LLM generalisation and diversity. In *International Conference on Learning Representations (ICLR)*, 2024.

[13] Jixuan Leng, Chengsong Huang, Banghua Zhu, and Jiaxin Huang. Taming overconfidence in LLMs: Reward calibration in RLHF. In *International Conference on Learning Representations (ICLR)*, 2025.

[14] Seungseop Lim, Gibaeg Kim, Hyunkyung Lee, Wooseok Han, Jean Seo, Jaehyo Yoo, and Eunho Yang. H-DDx: A hierarchical evaluation framework for differential diagnosis. 2025. NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[15] Che Liu, Haozhe Wang, Jiazhen Pan, Zhongwei Wan, Yong Dai, Fangzhen Lin, Wenjia Bai, Daniel Rueckert, and Rossella Arcucci. Beyond distillation: Pushing the limits of medical LLM reasoning with minimalist rule-based RL. 2025. Introduces the AlphaMed model; NeurIPS 2025 Workshop on Generative AI for Health (GenAI4Health).

[16] Mingyi Liu. The alignment tax: Response homogenization in aligned LLMs and its implications for uncertainty estimation, 2026.

[17] Zhiting Mei, Christina Zhang, Tenny Yin, Justin Lidard, Ola Shorinwa, and Anirudha Majumdar. Reasoning about uncertainty: Do reasoning models know when they don't know? In *Findings of the Association for Computational Linguistics: EACL 2026*, pages 3408–3458, Rabat, Morocco, 2026. Association for Computational Linguistics. doi: 10.18653/v1/2026.findings-eacl.178.

[18] Nicole Meister, Carlos Guestrin, and Tatsunori Hashimoto. Benchmarking distributional alignment of large language models. In *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pages 24–49, Albuquerque, New Mexico, 2025. Association for Computational Linguistics. doi: 10.18653/v1/2025.naacl-long.2.

[19] Changdae Oh, Qi Zeng, Qi Qi, Andrey Zhmoginov, Deren Lei, Yun He, Hoang Phan, Hangoo Kang, Azalia Mirhoseini, and Sharon Li. Sharpening tax in post-training. 2026.

[20] Isha Puri, Mehul Damani, Idan Shenfeld, Marzyeh Ghassemi, Jacob Andreas, and Yoon Kim. Escaping the mode: Multi-answer reinforcement learning in LMs. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 100127–100146. PMLR, 2026. arXiv version titled "Reaching Beyond the Mode: RL for Distributional Reasoning in Language Models".

[21] Team Olmo, Allyson Ettinger, Amanda Bertsch, Bailey Kuehl, David Graham, David Heineman, Dirk Groeneveld, Faeze Brahman, Finbarr Timbers, Hamish Ivison, et al. Olmo 3, 2025.

[22] Katherine Tian, Eric Mitchell, Allan Zhou, Archit Sharma, Rafael Rafailov, Huaxiu Yao, Chelsea Finn, and Christopher D. Manning. Just ask for calibration: Strategies for eliciting calibrated confidence scores from language models fine-tuned with human feedback. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2023.

[23] Xuezhi Wang, Jason Wei, Dale Schuurmans, Quoc Le, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *International Conference on Learning Representations (ICLR)*, 2023.

[24] Peter West and Christopher Potts. Base models beat aligned models at randomness and creativity. In *Conference on Language Modeling (COLM)*, 2025.

[25] Fang Wu, Weihao Xuan, Danny Xing, Ximing Lu, Mingjie Liu, Yi Dong, Zaid Harchaoui, and Yejin Choi. The invisible leash: Why RLVR may or may not escape its origin. 2025.

[26] Yang Yue, Zhiqi Chen, Rui Lu, Andrew Zhao, Zhaokai Wang, Yang Yue, Shiji Song, and Gao Huang. Does reinforcement learning really incentivize reasoning capacity in LLMs beyond the base model? In *Advances in Neural Information Processing Systems (NeurIPS 2025)*, 2025.

[27] Jiayi Zhang, Simon Yu, Derek Chong, Anthony Sicilia, Michael R. Tomz, Christopher D. Manning, and Weiyan Shi. Verbalized sampling: How to mitigate mode collapse and unlock LLM diversity. In *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, volume 306 of *Proceedings of Machine Learning Research*, pages 160073–160152. PMLR, 2026.